

Driver's License, Car Access, and Communication Apprehension

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Research or Analytical Question

Do people who have access to a car or a driver's license differ in communication apprehension from people who do not?

Answer, Response, + Summary of Results

I used responses from participants who could be matched across the Prolific and Qualtrics datasets. Participants without a driver's license or car access had higher average communication-apprehension scores, but none of the four differences were statistically significant.

Group-discussion apprehension was scored from Q1–Q6, and interpersonal apprehension was scored from Q13–Q18. Positively worded items were reverse scored. Each subscale ranged from 6 to 30, with higher scores indicating greater communication apprehension. A participant needed to answer all six items to receive a subscale score.

For driver's license, participants without a license had higher group-discussion apprehension ($M = 16.95$, $SD = 6.07$, $n = 20$) than those with a license ($M = 15.59$, $SD = 6.38$, $n = 132$), Welch's $t(25.79) = -0.93$, $p = .363$. They also had higher interpersonal apprehension ($M = 16.81$, $SD = 5.45$, $n = 21$) than those with a license ($M = 14.74$, $SD = 6.33$, $n = 129$), Welch's $t(29.54) = -1.58$, $p = .125$.

For car access, participants without access had higher group-discussion apprehension ($M = 17.00$, $SD = 5.37$, $n = 25$) than those with access ($M = 15.49$, $SD = 6.52$, $n = 126$), Welch's $t(39.45) = -1.24$, $p = .224$. They also had higher interpersonal apprehension ($M = 16.58$, $SD = 5.45$, $n = 26$) than those with access ($M = 14.64$, $SD = 6.36$, $n = 123$), Welch's $t(40.75) = -1.60$, $p = .118$.

Table 1

Comparison	Yes M (SD)	No M (SD)	Welch's t(df)	p
Driver's license — Group discussion	15.59 (6.38)	16.95 (6.07)	$t(25.79) = -0.93$	.363
Driver's license — Interpersonal	14.74 (6.33)	16.81 (5.45)	$t(29.54) = -1.58$	.125
Car access — Group discussion	15.49 (6.52)	17.00 (5.37)	$t(39.45) = -1.24$	.224
Car access — Interpersonal	14.64 (6.36)	16.58 (5.45)	$t(40.75) = -1.60$	.118

What questions or uncertainties remain?

- Ninety-nine of the 252 matched participants did not answer Q20 or Q21, so the results may not represent the full sample.

Analysis Code

```
import pandas as pd
from scipy import stats
import numpy as np

df = pd.read_csv("Summer2026DSHBDData/merged_dataset_both_scores.csv")

# group_discussion_prca: 18 - (Q2 + Q4 + Q6) + (Q1 + Q3 + Q5)
group_discussion_items = ["Q1_numeric", "Q2_numeric", "Q3_numeric",
                           "Q4_numeric", "Q5_numeric", "Q6_numeric"]
group_discussion_complete = df[group_discussion_items].notna().all(axis=1)
df["group_discussion_prca"] = np.where(group_discussion_complete,
    18 - df["Q2_numeric"] - df["Q4_numeric"] - df["Q6_numeric"]
    + df["Q1_numeric"] + df["Q3_numeric"] + df["Q5_numeric"], np.nan)
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# interpersonal_prca: 18 - (Q14 + Q16 + Q17) + (Q13 + Q15 + Q18)
interpersonal_items = ["Q13_numeric", "Q14_numeric", "Q15_numeric",
                       "Q16_numeric", "Q17_numeric", "Q18_numeric"]
interpersonal_complete = df[interpersonal_items].notna().all(axis=1)
df["interpersonal_prca"] = np.where(interpersonal_complete,
    18 - df["Q14_numeric"] - df["Q16_numeric"] - df["Q17_numeric"]
    + df["Q13_numeric"] + df["Q15_numeric"] + df["Q18_numeric"], np.nan)

def calculate_welch_ttest(df, group_col, score_col,
    group1="Yes", group2="No", exclude_value=None):
    if exclude_value:
        df_filtered = df[(df[group_col].notna()) &
            (df[group_col] != exclude_value)].copy()
    else:
        df_filtered = df[df[group_col].notna()].copy()
    df_filtered = df_filtered[df_filtered[score_col].notna()].copy()
    group1_data = df_filtered[df_filtered[group_col] == group1][score_col]
    group2_data = df_filtered[df_filtered[group_col] == group2][score_col]
    n1, n2 = len(group1_data), len(group2_data)
    mean1, std1 = group1_data.mean(), group1_data.std()
    mean2, std2 = group2_data.mean(), group2_data.std()
    t_stat, p_value = stats.ttest_ind(group1_data, group2_data,
        equal_var=False)

    var1, var2 = std1 ** 2, std2 ** 2
    df_welch = (var1 / n1 + var2 / n2) ** 2 / (
        (var1 / n1) ** 2 / (n1 - 1) + (var2 / n2) ** 2 / (n2 - 1))
    return {"yes_n": n1, "no_n": n2, "yes_mean": mean1, "yes_sd": std1,
        "no_mean": mean2, "no_sd": std2, "t_statistic": t_stat,
        "degrees_of_freedom": df_welch, "p_value": p_value}

def format_p(value):
    return f"{value:.3f}".lstrip("0")

results = [calculate_welch_ttest(df, "Q20", "group_discussion_prca"),
    calculate_welch_ttest(df, "Q20", "interpersonal_prca"),
    calculate_welch_ttest(df, "Q21", "group_discussion_prca",
        exclude_value="Not Sure"),
    calculate_welch_ttest(df, "Q21", "interpersonal_prca",
        exclude_value="Not Sure")]

display_data = [
    {"Comparison": "Driver's license - Group discussion",
        "Yes M (SD)": f"{results[0]['yes_mean']:.2f} ({results[0]['yes_sd']:.2f})",
        "No M (SD)": f"{results[0]['no_mean']:.2f} ({results[0]['no_sd']:.2f})",
        "Welch's t(df)": f"t({results[0]['degrees_of_freedom']:.2f}) = {results[0]['t_statistic']:.2f}",
        "p": format_p(results[0]['p_value'])},
    {"Comparison": "Driver's license - Interpersonal",
        "Yes M (SD)": f"{results[1]['yes_mean']:.2f} ({results[1]['yes_sd']:.2f})",
        "No M (SD)": f"{results[1]['no_mean']:.2f} ({results[1]['no_sd']:.2f})",
        "Welch's t(df)": f"t({results[1]['degrees_of_freedom']:.2f}) = {results[1]['t_statistic']:.2f}",
        "p": format_p(results[1]['p_value'])},
    {"Comparison": "Car access - Group discussion",
        "Yes M (SD)": f"{results[2]['yes_mean']:.2f} ({results[2]['yes_sd']:.2f})",
        "No M (SD)": f"{results[2]['no_mean']:.2f} ({results[2]['no_sd']:.2f})",
        "Welch's t(df)": f"t({results[2]['degrees_of_freedom']:.2f}) = {results[2]['t_statistic']:.2f}",
        "p": format_p(results[2]['p_value'])},
    {"Comparison": "Car access - Interpersonal",
        "Yes M (SD)": f"{results[3]['yes_mean']:.2f} ({results[3]['yes_sd']:.2f})",

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    "No M (SD)": f"{results[3]['no_mean']:.2f} ({results[3]['no_sd']:.2f})",
    "Welch's t(df)": f"t({results[3]['degrees_of_freedom']:.2f}) = {results[3]['t_statistic']:.2f}",
    "p": format_p(results[3]["p_value"])),
]

results_df = pd.DataFrame(display_data)
results_df.style.hide(axis="index").set_properties(**{"font-size": "9pt"})

```